National University of Sciences and Technology (NUST)
School of Electrical Engineering and Computer Science

Department of Computing

CS220: Database Systems

Class: BSAI – 2A

Lab 04: SQL JOINS

CLO-2: Formulate SQL queries to retrieve information from a relational database.

CLO 3: Build SQL (DDL &DML) queries to retrieve information from a relational database.

Date: 19 Februrary 2026

Time: 10:00 - 01:00

Lab 05: SQL JOINS

Introduction

A SQL join clause combines records from two or more tables in a database. It creates a set that can be saved as a table or used as it is. A JOIN is a means for combining fields from two tables by using values common to each.

Objectives

After performing this lab students should be able to: 1. Design SQL queries to retrieve data from multiple tables by using JOIN operation. 2. Select among INNER, OUTER, NATURAL, CROSS join as and when required.

Tools/Software Requirement

• MySQL Community Server 5.6
• MySQL Workbench 6.1
• Sakila Database

Description

1. This lab assumes that MySQL Community Server is running, **Sakila** database has been loaded using MySQL Workbench, and the query window is open.
2. Execute the JOIN queries on Sakila including cross join, inner join, natural join, outer join (left outer join, right outer join, and full outer join) and observe how each join operation differs from the other.

3. For your practice, create the following two tables and execute the following queries to better understand how each join operation works.

Users

id	name	course_id
1	Alice	1
2	Bob	1
3	Caroline	2
4	David	5
5	Emma	(NULL)

Course

course_id	name
1	HTML5
2	NULL
3	JavaScript
4	PHP
5	MySQL

a. Specify a primary-foreign key relationship between both tables.
b. Produce a set of all users who are enrolled in a course?
c. List of all students and their courses even if they're not enrolled in any course yet?
d. All records in both tables regardless of any match?

4. Try running following SQL JOIN queries. You can try equivalent SQL queries for the same result as well.

   a. List rental and return date for the movie BREAKING HOME (notice one of the join uses on clause where as the other uses using). 13 entries of renting this movie should be found from the database.

      use sakila;

      select r.rental_date, return_date
      from rental r **join** inventory i
      using (inventory_id) **join** film f
      on (f.film_id=i.film_id)
      where f.title **like** 'BREAKING HOME' **;**

   b. List of movie titles that were never rented. There are 43 of them. Notice that there are two outer joins as the movie might be missing altogether from the inventory but may still be included in the result set. On the other hand movie might be available in the inventory but never rented, the SQL query covers both cases.

      select f.title
      from film f **left join** inventory i
      using (film_id) **left join** rental r
      using (inventory_id)
      where (i.inventory_id **is null or** r.rental_id **is null);**

Lab Task

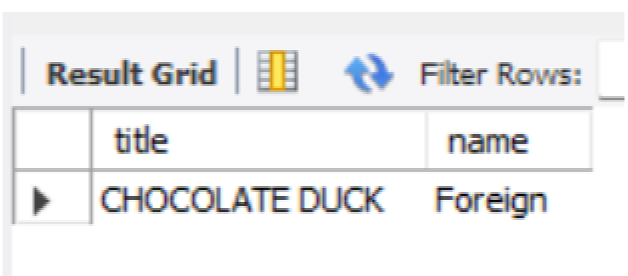
Write SQL queries for the following information needs. You should execute your attempts on SQL workbench and make necessary corrections if needed.

a. What category does the movie CHOCOLATE DUCK belong to?

CODE:

```
SELECT film.title, category.name
FROM film
JOIN film_category
ON film.film_id = film_category.film_id
Join category
ON film_category.category_id = category.category_id
WHERE film.title = "Chocolate Duck";
```

OUTPUT:



Result Grid | Filter Rows:

title	name
CHOCOLATE DUCK	Foreign

b. Track the location (city & country) of staff member JON.

CODE:

```
SELECT staff.first_name,country.country,city.city
From staff
Join address
On staff.address_id=address.address_id
Join city
On address.city_id=city.city_id
Join country
On city.country_id=country.country_id
Where staff.first_name ="JON";
```

OUTPUT:

Result Grid | Filter Rows:

first_name	country	city
Jon	Australia	Woodridge

c. Retrieve first and last name of actors who played in ALONE TRIP.

CODE:

```
SELECT first_name,last_name
From actor
Join film_actor
ON actor.actor_id=film_actor.actor_id
Join film
On film_actor.film_id=film.film_id
Where film.title="Alone Trip"
```

OUTPUT:

Result Grid | Filter Rows:

first_name	last_name
ED	CHASE
KARL	BERRY
UMA	WOOD
WOODY	JOLIE
SPENCER	DEPP
CHRIS	DEPP

d. List of movies in Games category having rental rate of more than $4 and sorted on movie titles.

CODE:

```
SELECT film.title
From film
Join film_category
On film_category.film_id=film.film_id
Join category
On film_category.category_id=category.category_id
Where rental_rate>"$4" and category.name= "Games"
Order by rental_rate
```

OUTPUT:

Result Grid Filter Rows:

title
HAUNTING PIANIST
ENCINO ELF
LADYBUGS ARMAGEDDON
FEATHERS METAL
CREATURES SHAKESPEARE
CURTAIN VIDEOTAPE
GRIT CLOCKWORK

e. Email addresses of customers who although rented a movie (include movie title in the output) but didn’t pay anything.

CODE:

```
SELECT customer.email, film.title
FROM customer
JOIN rental
  ON customer.customer_id = rental.customer_id
JOIN inventory
  ON rental.inventory_id = inventory.inventory_id
JOIN film
  ON inventory.film_id = film.film_id
LEFT JOIN payment
  ON rental.rental_id = payment.rental_id
WHERE payment.amount ="0";
```

OUTPUT:

email	title
HELEN.HARRIS@sakilacustomer.org	SMOKING BARBARELLA
CAROLYN.PEREZ@sakilacustomer.org	RIVER OUTLAW
CHRISTINE.ROBERTS@sakilacustomer.org	MOTIONS DETAILS
HEATHER.MORRIS@sakilacustomer.org	LAWLESS VISION
MILDRED.BAILEY@sakilacustomer.org	CHAMBER ITALIAN
TAMMY.SANDERS@sakilacustomer.org	TROUBLE DATE

f. List of unpaid rentals (film title, rental id, rental date and for how many days rented). List should be sorted on film title and rental date.

CODE:

```
SELECT film.title,rental.rental_id,rental.rental_date
FROM rental
JOIN inventory
On rental.inventory_id=inventory.inventory_id
Join film
On inventory.film_id=film.film_id
Left join payment
On rental.rental_id=payment.rental_id
Where payment.amount="0" OR payment.amount Is NULL
Order by film.title,rental.rental_date
```

OUTPUT:

title	rental_id	rental_date
BLADE POLISH	15497	2006-02-14 15:16:03
CHAMBER ITALIAN	14741	2006-02-14 15:16:03
CHASING FIGHT	12352	2006-02-14 15:16:03
CLEOPATRA DEVIL	14516	2006-02-14 15:16:03
CURTAIN VIDEOTAPE	15717	2006-02-14 15:16:03

g. How many films involve a word “Crocodile” and a “Shark”?

CODE:

```
Select count(*)
From film
Where film.description LIKE "%Crocodile%" and film.description LIKE "%Shark%"
```

OUTPUT:

Result Grid

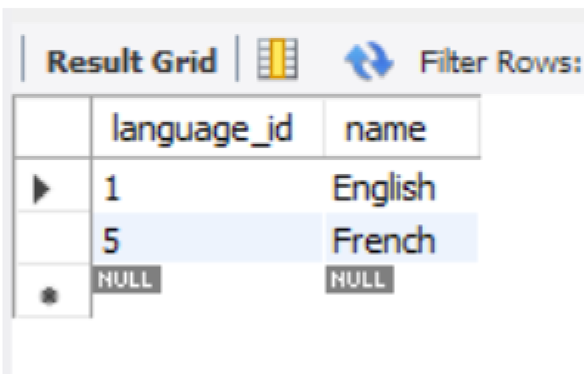
count(*)
10

h. Retrieve language ids of languages ENGLISH and FRENCH.

CODE:

```
Select language_id ,name
From language
Where name = "English" or name = "French"
```

OUTPUT:



language_id	name
1	English
5	French
NULL	NULL

Deliverables:

Submit a word document including the SQL queries to answer above-mentioned information needs as well as snapshot of their outcome when executed over MySQL using the Workbench.